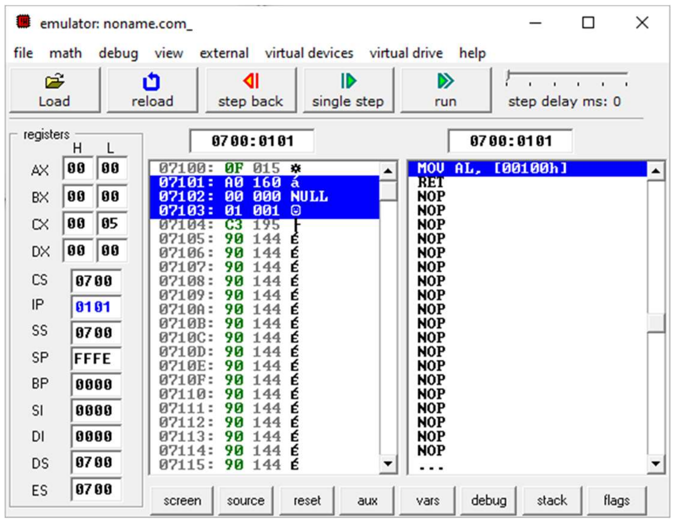
Name: Aetesam Asfar Ullah **Roll No.:** 02-134232-092

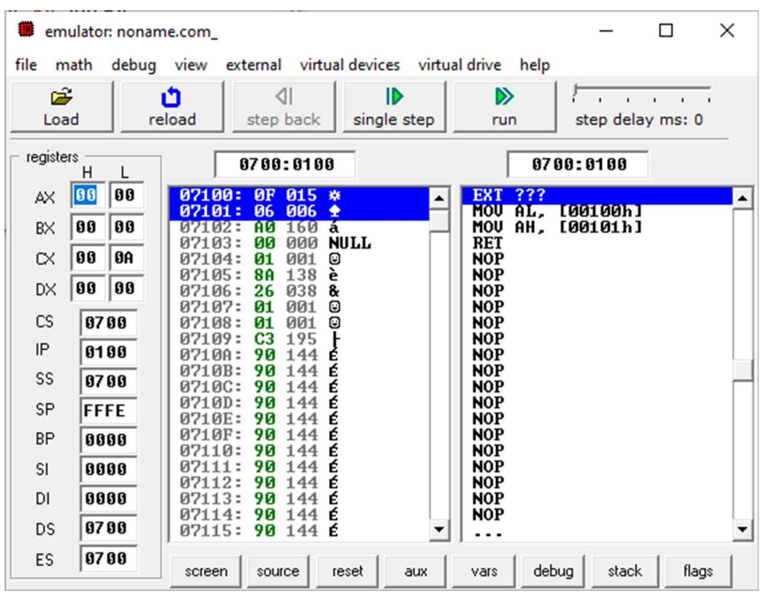
COAL LAB#4

Exercise 1:

Program 1:



Program 2:



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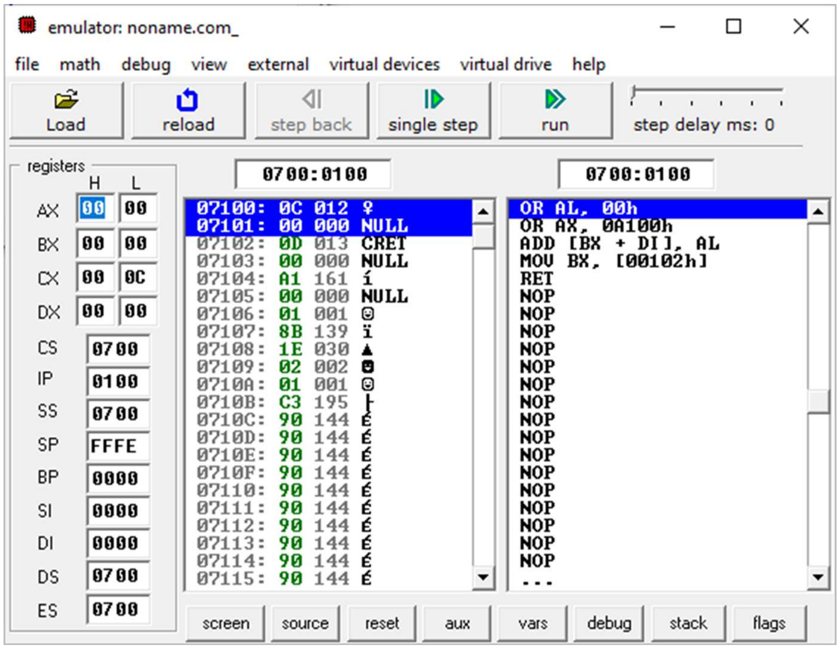
Program 3:



Exercise 2:

Register	Value	Value
DX	DH= 00	DL= 0F
DX	DH=06	DL=0F

Exercise 3:

```
org 100h
; Define two byte-size integers
num1 db 10d  ; num1 = 10
num2 db 20d  ; num2 = 20
; Add the two integers
mov al, num1
add al, num2
ret
```

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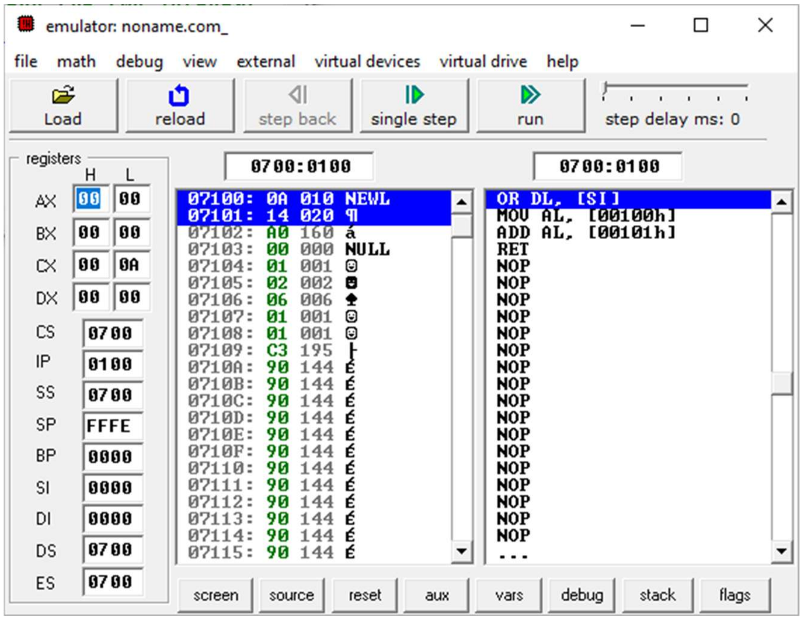
OUTPUT:



Exercise 4:

```
org 100h
; Define two byte-size integers
num1 db 30d  ; num1 = 30
num2 db 15d  ; num2 = 15
; Subtract num2 from num1
mov al, num1  ; Load num1 into AL
sub al, num2  ; Subtract num2 from AL
ret
```

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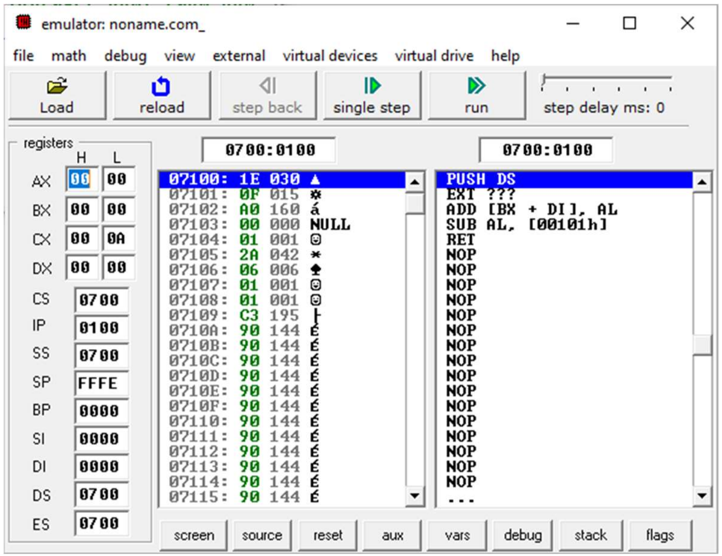
OUTPUT:



Exercise 5:

These are assembly directives used to define data in memory. Here is what each one stands for:

- **DB (Define Byte)**: Used to define a byte-sized data item (1 byte, 8 bits).
  - Example: "db 15d" defines a byte with the decimal value 15.
- **DW (Define Word)**: Used to define a word-sized data item (2 bytes, 16 bits).
  - Example: "dw 1000h" defines a word with the hexadecimal value 1000.
- **DD (Define Double Word)**: Used to define a double word-sized data item (4 bytes, 32 bits).
  - Example: "dd 12345678h" defines a double word with the hexadecimal value 12345678.
- **DQ (Define Quad Word)**: Used to define a quad word-sized data item (8 bytes, 64 bits).
  - Example: "dq 1234567890abcdefh" defines a quad word with the given hexadecimal value.
- **DT (Define Ten Bytes)**: Used to define a ten-byte-sized data item (10 bytes, 80 bits).
  - Example: "dt 1234567890abcdef0123456789h" defines a ten-byte value.